# Oluwatosin Ayeni

Senior Infrastructure Engineer (LLM Inference Systems) — Doubleword

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## PROFILE

Infrastructure Engineer specialising in LLM serving systems: batch inference throughput, request routing, and GPU cluster operations. Built and operated production Kubernetes LLM stacks achieving 5,965 tok/s peak on 2× H100 NVL with 100% SLO compliance at concurrency 128; KV-cache-aware routing, HPA autoscaling, Prometheus/Grafana/Loki/AlertManager. Profiled CUDA kernel bottlenecks for 3.8× throughput gains and 18× latency reduction.

## CORE COMPETENCIES

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|---|---|
| **Inference Infrastructure** | vLLM, SGLang, TensorRT-LLM, batch inference pipelines, request scheduling, KV-cache-aware load balancing, KEDA autoscaling |
| **Kubernetes & Platform** | Kubernetes (HPA, GPU node pools), Docker, Velero backup, Prometheus/Grafana/Loki/AlertManager, KEDA, GitHub Actions |
| **GPU & Performance** | CUDA kernel profiling (NVIDIA Nsight Systems/Compute), memory bandwidth, tensor parallelism, BFloat16/FP8, continuous batching |
| **Cloud & Serving** | RunPod, Vast.ai, LiteLLM gateway, FastAPI, multi-provider model routing, structured output |
| **Languages** | Python (primary), TypeScript, SQL; Rust (in progress — building inference tooling) |

## PROFESSIONAL EXPERIENCE

Jan 2025–present — **Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present — **Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present — **Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present — **AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 — **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

## KEY PROJECTS

**llm-inference-platform** — Production LLM serving on RunPod and Vast.ai: vLLM + SGLang + LiteLLM gateway, KEDA autoscaling, Prometheus/Grafana/Loki/AlertManager, Velero backup. GitHub

**vllm-k8s-stack** — Production Kubernetes LLM serving: KV-cache-aware routing, HPA autoscaling, Prometheus/Grafana. GitHub

**llm-inference-benchmark (H100)** — vLLM vs SGLang, H100 NVL — 5,965 tok/s peak, 100% SLO compliance. Live results

**inference-optimizer** — Production Inference Gateway: 3.87× faster, 14× throughput, 61.4% cost reduction. GitHub

## EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

## OPEN SOURCE

**GitHub** — 43 public repos — github.com/AyeniOluwatosinOlawale